

1. Prove the Gordon identity,

$$\bar{u}(p')\gamma^\mu u(p) = \bar{u}(p')\left[\frac{(p+p')^\mu}{2m} + iS^{\mu\nu}\frac{(p'-p)_\nu}{2m}\right]u(p)$$

You may want to start from the following (prove them first) :

$$\gamma^\mu \not{p} = \frac{1}{2}\{\gamma^\mu, \not{p}\} + \frac{1}{2}[\gamma^\mu, \not{p}] = -p^\mu - 2iS^{\mu\nu}p_\nu \quad (1)$$

$$\not{p}'\gamma^\mu = \frac{1}{2}\{\gamma^\mu, \not{p}'\} - \frac{1}{2}[\gamma^\mu, \not{p}'] = -p'^\mu + 2iS^{\mu\nu}p'_\nu \quad (2)$$

2. If $\bar{A} = \gamma^0 A^\dagger \gamma^0$ show $\bar{A} = A$ for the different choices: $A = \gamma^\mu, S^{\mu\nu}, i\gamma_5, \gamma^\mu\gamma_5, i\gamma_5 S^{\mu\nu}$.